Manish.instru@gmail.com Date: 14-12-2019 Saturday

LINEAR REGRESSION

By…ejagruti

Definition of a regression line:

A regression line is a straight line that describes how a response variable (y) changes as an explanatory variable (x) changes…

- Used to predict a y value given an x value.
- Requires an explanatory and a response variable.
- Given as an equation of a line in slope intercept form:

$$\hat{y} = a + bx$$

Read as: "y-hat"

a = y-intercept

b = slope

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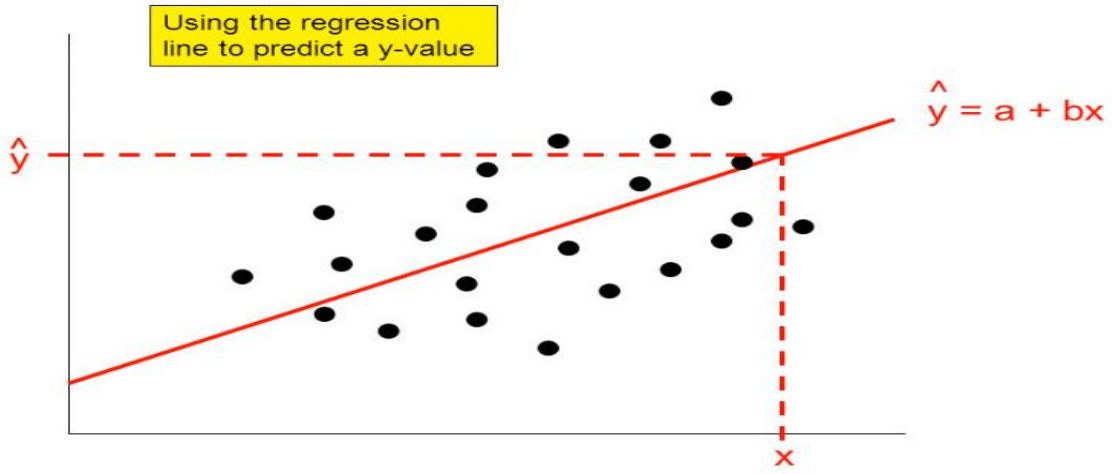
How It Works:



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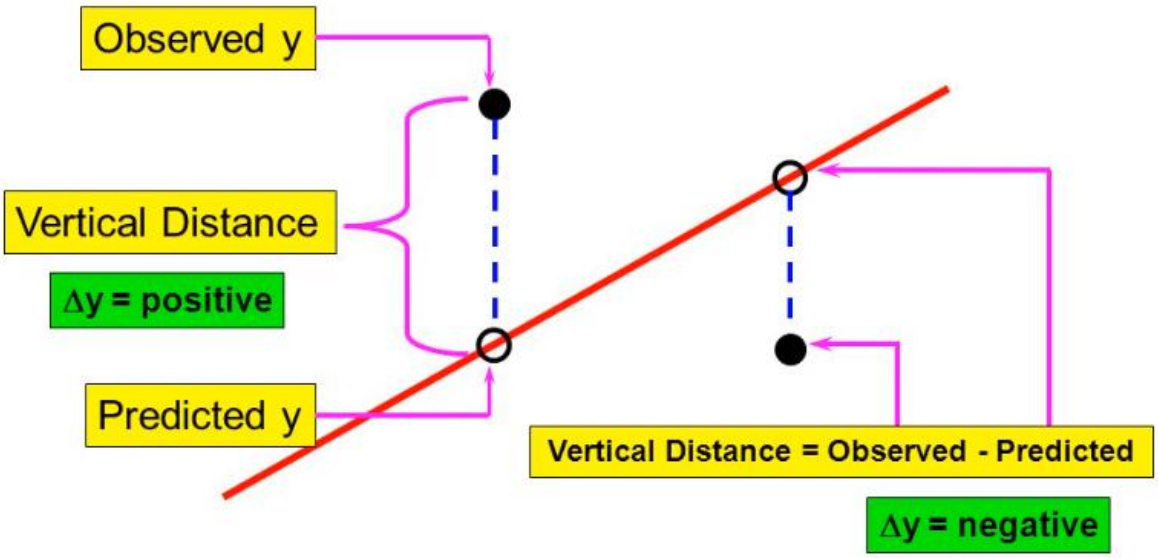
We are trying to find a line that minimizes the squares of the vertical distances



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Regression

Y Depends on X

$y = a + bx$

$[(y - \bar{y}) = b_{yx}(x - \bar{x})]$

X Depends on Y

$x = a + by$

$[(x - \bar{x}) = b_{xy}(y - \bar{y})]$

Where $\bar{y}$ - Mean of all y values

$\bar{x}$ - Mean of all x values

b_{yx} & b_{xy} are the Regression coefficient

① $b_{yx} = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$ | $b_{xy} = \dfrac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}$

② $b_{yx} = r\dfrac{\sigma_y}{\sigma_x}$ | $b_{xy} = r\dfrac{\sigma_x}{\sigma_y}$

$\sigma \rightarrow$ sigma

③ $b_{yx} = \dfrac{\text{Covariance}(x,y)}{(\sigma_x)^2}$ | $b_{yx} = \dfrac{\text{Covariance}(x,y)}{(\sigma_y)^2}$

$\text{cov}(x,y) = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{N}$ | $\text{cov}(x,y) = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{N-1}$